

Making Decisions: if / elif / else

The First Code That “Thinks” Before Acting

THE BIG QUESTION

**How do we make a program
actually behave differently
depending on what's true?**

What Is a Conditional?

CONDITIONAL

**RUNS A BLOCK OF CODE
ONLY IF A CONDITION IS TRUE**

Your First if Statement

- ```
is_raining = True

if is_raining:
 print("Bring an umbrella!")
```

```
is_raining = True

Bring an umbrella!
```

- ```
is_raining = False

if is_raining:
    print("Bring an umbrella!")
```

```
is_raining = False

(nothing prints)
```

Indentation: Not Just Style



```
if is_raining:
```

```
    print("Bring an umbrella!")
```



```
if is_raining:
```

```
print("Bring an umbrella!")
```

CORRECTLY INDENTED

```
Bring an umbrella!
```

MISSING INDENT

```
IndentationError:  
expected an indented block
```


else: The Otherwise

- ```
is_raining = False
if is_raining:
 print("Umbrella!")
else:
 print("Sunshine!")
```

```
is_raining = False

Sunshine!
```

- ```
is_raining = True
if is_raining:
    print("Umbrella!")
else:
    print("Sunshine!")
```

```
is_raining = True

Umbrella!
```


elif: More Than Two Paths

```
age = 15

if age < 13:
    print("Child ticket")
elif age < 20:
    print("Teen ticket")
elif age < 65:
    print("Adult ticket")
else:
    print("Senior ticket")
```

With age = 15 → "Teen ticket" prints. Change age and re-run to see the other branches.

PREDICT BEFORE YOU REVEAL

**score is 95 — that's true for BOTH
conditions below. Which message
prints?**

if score >= 60: "Pass" · elif score >= 90: "Pass with Distinction"

Only the First True Branch Runs

```
●  
score = 95  
  
if score >= 60:  
    print("Pass")  
elif score >= 90:  
    print("Pass with Distinction")
```

OUTPUT — the elif below is never even checked

Pass

Comparisons and Logic Inside a Condition

```
●  
age = 16  
is_student = True  
  
if age >= 13 and age <= 19 and is_student:  
    print("Eligible for the student teen discount!")
```

OUTPUT

Eligible for the student teen discount!

Common Structure Patterns



Algorithms Can Branch

ALGORITHM (Class 01 — imagined as a straight line)



ALGORITHMS CAN BRANCH (today — different paths for different data)

THE HERO DIAGRAM



Cyan = new today · DECISIONS sits right above RUN — it decides which version of the program actually happens

Your Profile Card Decides

Add an if/else using `is_student`,
and an if/elif/else chain using `age`
with at least three branches.

Change one value and re-run — confirm a different branch executes.

Profile Card Decides — Template

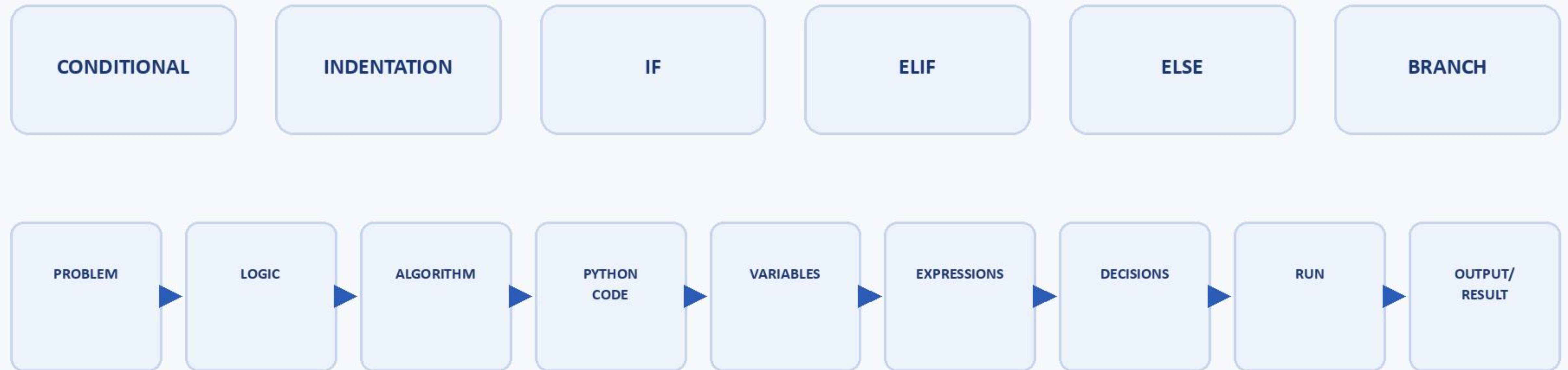
```
if is_student:  
    print("Time to study!")  
else:  
    print("Time to relax!")  
  
if age < 13:  
    print("Category: Child")  
elif age < 20:  
    print("Category: Teen")  
else:  
    print("Category: Adult")
```

REFLECT

**What changed when you edited
one variable and ran your
program again?**

A different branch executed — same code, new data.

Quick Recap



A QUESTION FOR NEXT TIME

Right now, every decision only runs once. What if a program needed to make the same decision — or repeat the same action — over and over?

FINAL TAKEAWAY

“Every program before today ran the same way, every single time. Today, for the first time, your code actually thinks before it acts — it looks at its own data and chooses a path.”